

Exact Causal Sources: Spectral Realization, Sharp Asymptotics, and Stability of Near-Optimal Seeds

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Abstract

For a finite controlled history tree, we construct one exact causal seed whose information is stochastically between the lower envelope of all adaptive transcript information laws and that envelope plus an independent exponential variable of mean $\log_2 e$. A refined stochastic comparison reduces the universal additive Shannon gap to $(1 + \log_2 e)/2$ and improves the simultaneous Renyi bounds. These comparisons yield positive-order Renyi approximation bounds, exact min-entropy, a constant gap to unconstrained coupling of all adaptive transcripts, and an exact Wasserstein identity for the entropy excess. For any fixed finite family of memoryless laws of common entropy h , we prove $C_n^{\text{causal}} = nh + \sqrt{2/\pi}(\sigma_{\max} - \sigma_{\min})\sqrt{n} + o(\sqrt{n})$, including zero lower variance. Every source with entropy within $o(\sqrt{n})$ of optimal has the same normalized information limit in W_1 . The greedy source additionally has convergence of all polynomial moments and asymptotically optimal fixed-confidence information quantiles. The square-root analysis uses explicit discrete Bellman barriers. A separately identified martingale theorem supplies the logarithmic upper bound when all positive varentropies agree. An irrational crash pair has an explicit finite-horizon error certificate. The literature map separates established coupling and nonlinear-CLT results from their compatible-tree realization and the consequences proved here.

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1 Position in the literature and the scope of the claims

This note concerns an exact, finite-horizon, history-indexed controlled interface. One random seed must implement the conditional output law at every history and for every requested action. The distinction between this problem and an unconstrained coupling of transcript laws is essential. All logarithms in information quantities are base two; $\ln$ is natural.

Ordinary minimum-entropy coupling. Kocaoglu, Dimakis, Vishwanath and Hassibi [1] study greedy minimum-entropy coupling in connection with entropic causal inference. Li [2] develops approximation methods for multiple marginals. Compton [3] proves the $\log_2 e$ additive guarantee for the ordinary greedy coupling algorithm. Compton et al. [4] improve the Shannon guarantees to $\log_2(e)/e$ for two marginals and $(1 + \log_2 e)/2$ for arbitrarily many marginals, using their profile method. These results concern fixed marginal distributions. Their algorithms do not, by their statements alone, provide a compatible decoder at shared history-action nodes of a controlled tree. The tree construction below supplies that additional compatibility directly. Its initial $\log_2 e$ bound is not asserted to improve those ordinary-coupling constants. We subsequently transfer the residual-mass refinement of [4] to the tree construction, recovering the $(1 + \log_2 e)/2$ Shannon bound and a stronger stochastic sandwich.

Information-spectrum converses and Renyi entropy. Shkel and Yadav [5], Theorem 1 and Section III-B, give information-spectrum lower bounds for functional representations and minimum Shannon/Renyi entropy couplings. The lower envelope used here is an application of that spectrum viewpoint to all deterministic adaptive-policy transcript laws. The converse is reproved below. Ma, Wang and Wu [6] establish parameter-dependent greedy approximation guarantees for ordinary two-marginal minimum-Renyi coupling; their error tends to zero as the order tends to infinity. Consequently neither “Renyi coupling” nor “a vanishing high-order approximation gap” should be advertised as new in itself. Our bound is derived from one compatible tree seed, simultaneously for all orders, with an explicit elementary error function. The recent extreme-point work of Ma et al. [7] addresses exact ordinary coupling through a different finite-dimensional route.

Controlled central limits. Peng’s nonlinear central limit theory [8] and subsequent quantitative work [9] provide the established setting of uncertain conditional variance and nonlinear heat equations. Peng and Zhou [10], Theorem 2, give an explicit one-sided tail capacity. With lower and upper standard deviations b, a , its complement is precisely the CDF $F_{a,b}$ used below. We do not claim that this profile or the underlying extremal-variance principle is new. We give a direct discrete comparison for arbitrary finite increment laws and then transfer it to the information of an exact causal seed. The CDF envelope defines an ordinary probability law, but this must not be confused with representing the entire nonlinear expectation by a single probability measure. Different thresholds can have different optimizing policies; when $a > b$, the envelope has positive mean even though every individual centered transcript is a martingale sum.

Equal conditional variances. The only substantive external estimate used in the proofs is the nonuniform martingale Berry–Esseen inequality of Fan, Grama and Liu [11], Theorem 2.1. We state exactly the specialization needed and check its hypotheses. Its unnamed absolute constant is retained as C_{FGL} ; we do not assign it an unproved numerical value. All other arguments below are given directly.

What this manuscript establishes, and what it does not. The central finite-tree result is a stochastic sandwich for the information of one compatible greedy seed. Its consequences include simultaneous Renyi bounds, exact min-entropy, a constant gap to the ordinary coupling

of *all adaptive* transcript laws, and a Wasserstein stability identity. The latter forces a common information limit for every first-order near-optimal seed on the square-root scale. We also prove convergence of all polynomial moments for the greedy construction and explain why that stronger conclusion fails for arbitrary near-optimal seeds. The literature map checks the primary sources cited above, including 2025 work; it is not an exhaustive priority certificate. We make no claim to have discovered a new G -normal law or to have settled the sharp logarithmic regime.

2 Model and the sharp square-root statement

Fix a finite nonempty action family $\mathcal{A} = \{1, \dots, m\}$. Action i requests a finite law P_i on its own finite alphabet $\mathcal{Y}_i$; these alphabets need not have equal cardinalities. Zero-probability symbols are removed. Assume in the asymptotic sections that

$$H(P_i) = h \quad (i \in \mathcal{A}), \quad \sigma_i^2 = \text{Var}_{P_i}(-\log_2 P_i(Y)), \quad a = \max_i \sigma_i, \quad b = \min_i \sigma_i.$$

A seed S is sampled once, independently of external policy randomness. At round t , the output is a deterministic function of S , the complete past action-output history, and the currently requested action. Under every adaptive policy its conditional law must be the requested P_i . Let C_n^{causal} be the infimum Shannon entropy of finite seeds with this property. Seeds may depend on n . Responses to the same action at distinct histories need not agree. Thus this is a history-indexed causal model, not the stronger synchronous model in which a time-action coordinate must be shared across different histories.

Theorem 1. *For every fixed finite family as above,*

$$C_n^{\text{causal}} = nh + \sqrt{\frac{2}{\pi}}(a - b)\sqrt{n} + o(\sqrt{n}). \quad (1)$$

If $a = b = 0$, then $C_n^{\text{causal}} = nh$. If $a = b > 0$, then

$$0 \leq C_n^{\text{causal}} - nh = O(\log(n + 1)). \quad (2)$$

Write ν_i for the law of $D_i = -\log_2 P_i(Y) - h$, so $\mathbb{E}D_i = 0$. Define

$$(Tf)(x) = \min_{i \in \mathcal{A}} \mathbb{E}f(x - D_i), \quad G_0(x) = \mathbf{1}_{\{x \geq 0\}}, \quad G_n = T^n G_0. \quad (3)$$

Each G_n is a finite-support CDF; let $Z_n \sim G_n$, $e_n = \mathbb{E}Z_n$. Here Z_n is centered relative to the entropy baseline; the uncentered information envelope below has law $nh + Z_n$. For a deterministic policy π , let $M_{\pi,n}$ be transcript information minus nh . Backward induction gives

$$G_n(x) = \min_{\pi} \mathbb{P}(M_{\pi,n} \leq x). \quad (4)$$

After the first action and observed output, a separate continuation may be chosen for each branch and remaining threshold. This gives precisely (3). Extra observed labels with the same increment do not change the optimal value: each continuation has the same available kernels and threshold. A randomized policy gives a mixture of deterministic-policy values after conditioning on its private randomness, so cannot lower the minimum further. Deterministic actions contribute no additional information beyond their preceding history.

The proof will establish

$$0 \leq C_n^{\text{causal}} - nh - e_n \leq \log_2 e, \quad e_n/\sqrt{n} \longrightarrow \sqrt{2/\pi}(a - b). \quad (5)$$

3 Global realization with a constant entropy gap

We prove a finite-tree statement, also valid for history-dependent kernels and any finite number of actions. At history v , action A has its own nonempty finite output alphabet $\mathcal{Y}(v, A)$ and kernel $K_v(\cdot \mid A)$ normalized on that alphabet. Neither a common alphabet nor equal alphabet sizes are assumed; zero-probability outputs may be omitted. All sums over y below are over $\mathcal{Y}(v, A)$. A history is v , and its child after action A and output y is vAy . Set

$$p(\emptyset) = 1, \quad p(vAy) = p(v)K_v(y \mid A).$$

These cylinder weights satisfy $\sum_y p(vAy) = p(v)$ for each action separately. For a policy π , let $\mathcal{L}_\pi$ denote its terminal histories. Their weights sum to one and form its transcript law. Define $I_\pi = -\log_2 p(T_\pi)$ and

$$F_*(t) = \min_{\pi} \mathbb{P}(I_\pi \leq t), \quad Z \sim F_*, \quad L = \mathbb{E}Z.$$

There are finitely many deterministic response maps on this finite tree. Any seed can be coarsened by merging atoms with the same response map, without changing its outputs or increasing a positive-order Renyi entropy. The feasible weights on these maps form a nonempty compact polytope: nonemptiness follows, for example, by independently pre-sampling a response at every history-action node. Positive-order Renyi entropies, Shannon entropy, and min-entropy are continuous on this simplex, so their minima are attained. We continue to use infimum notation when convenient.

Lemma 2 (Finite global realization). *There is a finite exact causal seed S_G , supported on at most the number of terminal histories in the full tree, such that*

$$L \leq C \leq H(S_G) \leq L + \log_2 e. \quad (6)$$

Proof. A residual flow r is nonnegative and satisfies $\sum_y r(vAy) = r(v)$ for every action. A deterministic response strategy d chooses an output at each history-action pair. Its compatibility indicator $\chi_d(v)$ equals one when all its outputs on the path to v agree with v . It is a flow of root mass one.

For any residual flow compute from the leaves upward

$$B_r(v) = r(v) \quad (v \text{ terminal}), \quad B_r(v) = \min_A \max_y B_r(vAy) \quad (v \text{ nonterminal}).$$

There are two elementary consequences. First, the largest extractable strategy weight is $B_r(\emptyset)$: for each possible action the strategy selects a child of maximal bottleneck, and this is done recursively. Nonnegativity on terminal histories implies nonnegativity at all histories by flow conservation. Second, a policy choosing a minimizing action at every history has

$$r(v) \leq B_r(\emptyset) \quad (v \in \mathcal{L}_{\pi_r}).$$

Indeed, along every output of that policy, bottleneck values never increase. This policy also proves that no larger strategy weight is possible. If the root mass is positive, every action has a positive child, and induction gives a positive root bottleneck.

Starting from $r = p$, repeatedly extract a maximizing strategy of weight $w = B_r(\emptyset)$ and replace r by $r - w\chi_d$. At least one positive terminal residual becomes zero at each extraction; otherwise the same strategy would permit a larger weight. Residuals never increase, so the algorithm terminates after finitely many steps. Its weights w_j are nonincreasing, sum to one, and satisfy

$$p(v) = \sum_j w_j \chi_{d_j}(v). \quad (7)$$

Choosing strategy d_j with probability w_j gives all the prescribed history weights simultaneously. Ratios of child to parent weights give the correct conditional laws at every positive-probability history. Thus this is one exact causal seed for all policies.

Fix $0 < \delta \leq 1$ and stop just after all weights larger than δ have been extracted. The remaining root mass is $r(\emptyset) = \sum_{j:w_j \leq \delta} w_j$. Its bottleneck is at most δ . The minimizing policy above therefore gives

$$\begin{aligned} \sum_{j:w_j \leq \delta} w_j &\leq \max_{\pi} \sum_{v \in \mathcal{L}_{\pi}} \min\{p(v), \delta\} \\ &= \max_{\pi} \mathbb{E} \min\{1, \delta 2^{I_{\pi}}\} \leq \mathbb{E} \min\{1, \delta 2^Z\}. \end{aligned} \quad (8)$$

The last inequality uses stochastic dominance: $F_* \leq F_{I_{\pi}}$. The zero residual case satisfies the same inequality trivially. For $J = -\log_2 w_{S_G}$, substitute $\delta = 2^{-t}$ to obtain

$$\mathbb{P}(J \geq t) \leq \mathbb{E} \min\{1, 2^{Z-t}\} = \mathbb{P}(Z + E \geq t),$$

where E is independent exponential of rate $\ln 2$. Integrating these nonnegative-variable tails yields $H(S_G) \leq L + 1/\ln 2$.

For the converse, under any deterministic policy each seed atom is contained in its transcript atom. Its surprisal therefore dominates the transcript surprisal under the induced coupling. Hence $F_S \leq F_{I_{\pi}}$ for all π , and thus $F_S \leq F_*$. Integration gives $H(S) \geq L$. This proves (6). $\square$

Identity case: the gap can vanish exactly. Suppose that at time t every action requests the same law R_t , independently of the preceding history; action-specific bijective relabelings of its support are also allowed. Sample independent $U_t \sim R_t$ once and use U_t , with the requested relabeling, at time t . This is an exact causal seed of entropy $\sum_{t=1}^n H(R_t)$. Under every deterministic policy, the transcript recovers precisely $U_1, \dots, U_n$, so all transcript information laws coincide with $-\log_2 \mathbb{P}(U_1, \dots, U_n)$. Therefore

$$C = L = \sum_{t=1}^n H(R_t).$$

In particular, if all actions request a fixed law P , then $C_n^{\text{causal}} = nH(P)$, without a uniformity assumption on P . Thus the universal additive bound is not a positive cost incurred by every instance.

Refinement: the constant $\log_2 e$ is not sharp. The same tree algorithm satisfies the stronger bound

$$L \leq C \leq H(S_G) \leq L + c_*, \quad c_* := \frac{1 + \log_2 e}{2} = 1.2213475204 \dots$$

In fact, let A be a fair Bernoulli variable, let $E \sim \text{Exp}(\ln 2)$, and let A, E, Z be independent. Then

$$Z \preceq_{\text{st}} J_{S_G} \preceq_{\text{st}} Z + W, \quad W = A(1 + E). \quad (\text{R})$$

The value c_* and the residual-mass refinement below are known for ordinary greedy coupling from Compton et al. [4], Theorem 4.2 and Appendix C. The following argument verifies their transfer to compatible history trees and expresses the refinement as a stochastic comparison. We claim no priority for the numerical constant itself.

Proof. Use the residual and cutoff δ from Lemma 2. On its minimizing policy, every terminal residual satisfies $r(v) \leq \delta$. If $p(v) > \delta$, that leaf must already have been visited by an extracted strategy: otherwise $r(v) = p(v) > \delta$. Every previously extracted weight exceeds δ . Thus, writing $p = p(v)$,

$$r(v) \leq a_\delta(p) := \begin{cases} p, & p \leq \delta, \\ \min\{\delta, p - \delta\}, & p > \delta. \end{cases}$$

For $p > \delta$, the latter minimum is at most $p/2$. Define the nondecreasing function

$$\psi(u) = \begin{cases} u, & 0 \leq u \leq \frac{1}{2}, \\ \frac{1}{2}, & \frac{1}{2} < u < 1, \\ 1, & u \geq 1. \end{cases}$$

For positive p we have $a_\delta(p) \leq p\psi(\delta/p)$; zero-weight leaves contribute zero. Summing over the minimizing policy and using stochastic dominance of Z yields

$$\mathbb{P}(J_{SG} \geq t) \leq \mathbb{E}\psi(2^{Z-t}) = \mathbb{P}(Z + W \geq t), \quad t \geq 0.$$

Indeed, W has mass $1/2$ at zero, no mass in $(0, 1)$, and survival probability 2^{-s} for $s \geq 1$. At $s = 0$ its inclusive survival is one; for $0 < s < 1$ it is $1/2$. These are exactly the three pieces above. Taking thresholds decreasing to t gives the strict-tail inequalities and hence (R), including its atoms. The lower comparison is unchanged. Taking expectations gives $\mathbb{E}W = (1 + \log_2 e)/2$. Since J_{SG} is bounded and $Z + W$ has an unbounded upper tail, its entropy bound is actually strict for each individual finite tree; a supremum over trees can still equal c_* . $\square$

Consequences and the remaining sharpness question. In particular, the additive $\log_2 e$ in any Shannon-gap estimate below may be replaced by c_* . The earlier exponential sandwich remains valid because $W \preceq_{\text{st}} E$; we retain it where it gives simpler formulas. There is also a simultaneous improvement of the positive-order Renyi errors in Theorem 4:

$$\tilde{g}_\alpha = \frac{\log_2(\frac{1}{2} + 2^{-\alpha}/\alpha)}{1 - \alpha} \quad (\alpha > 0, \alpha \neq 1), \quad \tilde{g}_1 = c_*.$$

To check it, independence gives $\mathbb{E}2^{(1-\alpha)W} = \frac{1}{2} + 2^{-\alpha}/\alpha$. Apply (R) to the increasing exponential for $\alpha < 1$ and to the decreasing exponential for $\alpha > 1$, reversing the sign again when dividing by $1 - \alpha$. The result is $R_\alpha(Z) \leq C_\alpha \leq H_\alpha(S_G) \leq R_\alpha(Z) + \tilde{g}_\alpha$. For example, $\tilde{g}_2 = \log_2(8/5)$, whereas the original error is $g_2 = 1$.

For an individual tree write $\Delta_{\text{opt}} = C - L$ and $\Delta_G = H(S_G) - L$. Let K_{opt} be the supremum of the first quantity over all finite controlled trees, and K_G the supremum of the second over these trees and permitted greedy tie choices. The proved bounds are

$$\boxed{\frac{1}{3} \leq K_{\text{opt}} \leq K_G \leq c_* < \log_2 e.}$$

Here is an elementary exact lower example. Take one round and two actions, uniform on two and three symbols, respectively. Its envelope is the constant $Z = \log_2 3$. Any seed can be coarsened to its pair of outputs, so its minimum entropy is that of a 2×3 coupling matrix. Write its first row as (a_1, a_2, a_3) , with $0 \leq a_i \leq 1/3$ and $\sum_i a_i = 1/2$; the second row is $1/3 - a_i$. Entropy is

concave on this compact polytope and attains a minimum at a vertex. Every vertex has, up to permutation, first row $(1/3, 1/6, 0)$: at least two coordinates must be at bounds. Its nonzero entries are $(1/3, 1/3, 1/6, 1/6)$, giving $C = \log_2 3 + 1/3$. Greedy has the same weights. This supplies the displayed lower bound, not an assertion that $1/3$ is the best lower example in the ordinary-coupling literature. Neither K_{opt} nor K_G is identified exactly here; the established conclusion is that the original $\log_2 e$ constant is not sharp.

In the equal-entropy memoryless model, $F_*(t) = G_n(t - nh)$ by (4). Lemma 2 proves the first inequality in (5). In particular, realizability introduces no error proportional to the number of rounds or blocks.

4 Finite-tree consequences: all Renyi orders and stability

For an exact seed S , write $J_S = -\log_2 \mathbb{P}(S)$. The envelope variable in Lemma 2 is denoted by Z and need not itself be the information of any discrete source. Its law is nevertheless a well-defined finite-support probability distribution.

Theorem 3 (Stochastic realization). *For every finite controlled history tree and its greedy seed S_G ,*

$$Z \preceq_{\text{st}} J_{S_G} \preceq_{\text{st}} Z + E, \quad E \sim \text{Exp}(\ln 2), \quad E \perp Z. \quad (9)$$

Every exact seed satisfies $Z \preceq_{\text{st}} J_S$. The notation $X \preceq_{\text{st}} Y$ means $F_X(t) \geq F_Y(t)$ for all t .

Proof. The atom-containment argument in Lemma 2 proves the lower stochastic bound for every seed. Its inequality (8), with $\delta = 2^{-t}$, gives $\mathbb{P}(J_{S_G} \geq t) \leq \mathbb{E} \min(1, 2^{Z-t}) = \mathbb{P}(Z + E \geq t)$ for $t \geq 0$. All variables are nonnegative, so the comparison also holds for $t < 0$. Taking one-sided limits supplies the CDF formulation, including at atoms. $\square$

This is an ordering of distributions. It does not say that the algorithm constructs $J_{S_G} = Z + E$, or that a seed's excess is an independent exponential variable on its operational probability space.

For $0 < \alpha < \infty$, $\alpha \neq 1$, define

$$H_\alpha(S) = \frac{1}{1-\alpha} \log_2 \sum_s \mathbb{P}(S = s)^\alpha, \quad R_\alpha(Z) = \frac{1}{1-\alpha} \log_2 \mathbb{E} 2^{(1-\alpha)Z}.$$

Set $H_1 = H$, $R_1(Z) = \mathbb{E}Z$, and let C_α be the infimum of H_α over exact causal seeds for this tree. Write

$$g_\alpha = \frac{\log_2 \alpha}{\alpha - 1} \quad (\alpha \neq 1), \quad g_1 = \log_2 e.$$

Theorem 4 (A single seed for every positive Renyi order). *Simultaneously for all $0 < \alpha < \infty$,*

$$R_\alpha(Z) \leq C_\alpha \leq H_\alpha(S_G) \leq R_\alpha(Z) + g_\alpha. \quad (10)$$

The construction of S_G is independent of α .

Proof. Put $M_\alpha = \mathbb{E} 2^{(1-\alpha)Z}$. Independence and direct integration give

$$\mathbb{E} 2^{(1-\alpha)(Z+E)} = M_\alpha / \alpha, \quad \mathbb{E} 2^{(1-\alpha)E} = \int_0^\infty (\ln 2) e^{-\alpha(\ln 2)t} dt = 1/\alpha.$$

For $0 < \alpha < 1$, the function $2^{(1-\alpha)x}$ is increasing, so (9) yields $M_\alpha \leq \sum_s \mathbb{P}(S_G = s)^\alpha \leq M_\alpha / \alpha$. Take logarithms and divide by $1 - \alpha > 0$. For $\alpha > 1$, the moment inequalities reverse; division by $1 - \alpha < 0$ reverses them again, giving the same bound. For arbitrary S , the lower stochastic bound proves the lower entropy bound by the same sign check. The case $\alpha = 1$ is Lemma 2, or follows by taking limits. $\square$

Theorem 5 (Exact min-entropy). *Let B_p be the bottleneck recursion in Lemma 2. Then*

$$C_\infty = H_\infty(S_G) = -\log_2 B_p(\emptyset).$$

For memoryless laws $P_1, \dots, P_m$ repeated for n rounds, without any equal-entropy assumption,

$$C_{n,\infty} = n \max_i H_\infty(P_i). \quad (11)$$

Proof. An atom of any exact seed determines a pure response strategy. Its weight cannot exceed that strategy's extractable weight, and hence cannot exceed $B_p(\emptyset)$. The first greedy atom has exactly that weight, and all later weights are no larger. This proves the first claim. Put $\beta = \min_i \max_y P_i(y)$. Induction on remaining depth shows $B_p(v) = p(v)\beta^{n-|v|}$ in the memoryless model. Taking minus base-two logarithms proves (11). $\square$

No finite bound at order $\alpha = 0$ follows by sending $\alpha \downarrow 0$ in (10): g_α diverges. The number of atoms is instead bounded by the number of full-tree terminal histories.

Theorem 6 (Cost of compatibility with all adaptive transcripts). *Let M_α denote the minimum Renyi entropy of an ordinary joint coupling of the transcript laws for all deterministic adaptive policies in the finite tree. Then*

$$0 \leq C_\alpha - M_\alpha \leq g_\alpha \quad (0 < \alpha < \infty), \quad C_\infty = M_\infty.$$

Proof. An exact causal seed induces a joint vector of transcripts by running each deterministic policy on that same seed. This vector is a deterministic function of the seed. Merging probability atoms cannot increase any positive-order Renyi entropy, so $M_\alpha \leq C_\alpha$. Conversely, the information of any ordinary joint coupling dominates the information of each of its coordinates. It therefore dominates Z stochastically and has entropy at least $R_\alpha(Z)$. Theorem 4 now gives $C_\alpha \leq R_\alpha(Z) + g_\alpha \leq M_\alpha + g_\alpha$. For the infinite-order endpoint, the largest atom in any such coupling is at most $\min_\pi \max_{v \in \mathcal{L}_\pi} p(v)$. The minimizing bottleneck policy and the maximizing response strategy prove that this min-max is $B_p(\emptyset)$; apply Theorem 5. $\square$

All adaptive policies are required here. Replacing them by constant policies, or by one-time output marginals, changes the lower bound. This is a finite-horizon comparison, not a claim that a single seed works for every horizon.

Corollary 7 (Tail and quantile guarantees). *For all real t and $s \geq 0$,*

$$\mathbb{P}(J_{S_G} > t + s) \leq \mathbb{P}(Z > t) + 2^{-s}.$$

Writing $q_p(X) = \inf\{x : F_X(x) \geq p\}$, for $0 < \delta < \epsilon < 1$,

$$q_{1-\epsilon}(Z) \leq q_{1-\epsilon}(J_{S_G}) \leq q_{1-\epsilon+\delta}(Z) + \log_2(1/\delta).$$

Proof. Use (9) and $\{Z + E > t + s\} \subset \{Z > t\} \cup \{E > s\}$. Substitute $t = q_{1-\epsilon+\delta}(Z)$ and $s = \log_2(1/\delta)$ for the upper quantile bound. Stochastic dominance gives the lower one. Finite support ensures the defining CDF levels are attained. $\square$

Lemma 8 (Exact transport identity). *For every finite-entropy exact seed S ,*

$$W_1(\mathcal{L}(J_S), \mathcal{L}(Z)) = H(S) - \mathbb{E}Z. \quad (12)$$

In particular, if $H(S) \leq C_1 + \Delta$, then this distance is at most $\Delta + \log_2 e$.

Proof. For any coupling, $\mathbb{E}|J_S - Z| \geq |\mathbb{E}J_S - \mathbb{E}Z|$. For a uniform $U \in (0, 1)$, stochastic dominance gives $q_U(J_S) \geq q_U(Z)$. This quantile coupling attains the lower bound: its expected absolute difference is $\mathbb{E}J_S - \mathbb{E}Z$. The last assertion follows from Lemma 2. $\square$

Theorem 9 (Transfer of any diverging-scale envelope limit). *Consider a sequence of finite trees, envelope variables Z_n , centers m_n , and scales $r_n \rightarrow \infty$. Suppose $\mathcal{L}((Z_n - m_n)/r_n) \rightarrow \mu$ in W_1 . Every exact seed sequence satisfying $H(S_n) - C_{n,1} = o(r_n)$ then obeys*

$$\mathcal{L}((J_{S_n} - m_n)/r_n) \longrightarrow \mu \quad \text{in } W_1.$$

Proof. Translation invariance, scaling, and Lemma 8 show that the distance between the two normalized information laws is bounded by $(H(S_n) - C_{n,1} + \log_2 e)/r_n \rightarrow 0$. Apply the triangle inequality. $\square$

5 An explicit finite-horizon Bellman comparison

Assume for now $a \geq b > 0$. Let Φ be the standard normal CDF and set

$$F(x) = \begin{cases} \frac{2b}{a+b}\Phi(x/b), & x \leq 0, \\ 1 - \frac{2a}{a+b}(1 - \Phi(x/a)), & x > 0. \end{cases} \quad (13)$$

This is the complement of the one-sided tail capacity in Peng-Zhou [10], Theorem 2. We verify the profile directly. Define

$$K = \frac{\sqrt{2/\pi}}{a+b}, \quad m_3 = \max_A \mathbb{E}|D_A|^3, \quad \mathcal{C} = K \left(\frac{m_3}{3b^2} + a \right), \quad u_t(x) = F(x/\sqrt{t}) \quad (t > 0).$$

Lemma 10 (Explicit comparison). *For every $\tau \geq 1$, $R \geq 0$, and integer $n \geq 0$, put*

$$\eta = \max\{F(-R/\sqrt{\tau}), 1 - F(R/\sqrt{\tau})\}.$$

Then, for every $x \in \mathbb{R}$,

$$\boxed{\begin{aligned} F\left(\frac{x-R}{\sqrt{n+\tau}}\right) - \eta - \frac{3\mathcal{C}}{\sqrt{\tau}} &\leq G_n(x), \\ G_n(x) &\leq F\left(\frac{x+R}{\sqrt{n+\tau}}\right) + \eta + \frac{3\mathcal{C}}{\sqrt{\tau}}. \end{aligned}} \quad (14)$$

Proof. For x of either sign let $c = b$ on the negative half-line and $c = a$ on the positive half-line. Direct differentiation gives

$$F'(x) = K e^{-x^2/(2c^2)}, \quad F''(x) = -\frac{Kx}{c^2} e^{-x^2/(2c^2)}.$$

The first two derivatives extend continuously across zero; in particular $F''(0) = 0$. The one-sided third derivatives satisfy

$$|F'''(x)| = \frac{K}{c^2} \left| \frac{x^2}{c^2} - 1 \right| e^{-x^2/(2c^2)} \leq \frac{2K}{b^2}.$$

Consequently F'' is globally Lipschitz with that constant. Taylor's theorem therefore applies even when an increment crosses the interface.

The function u_t satisfies the following identity everywhere, including $x = 0$:

$$\partial_t u_t(x) = \frac{1}{2} \min\{a^2 \partial_{xx} u_t(x), b^2 \partial_{xx} u_t(x)\}. \quad (15)$$

The minimum over i in the quadratic term uses only the extreme variances. On $x < 0$ the second derivative is positive and the minimum selects b^2 ; on $x > 0$ it is negative and selects a^2 . On each side both expressions equal $-x \partial_x u_t(x)/(2t)$.

Because $\mathbb{E}D_A = 0$, the spatial Taylor remainder gives

$$|Tu_t - u_t - \partial_t u_t| \leq \frac{Km_3}{3b^2} t^{-3/2} \quad \text{uniformly in } x.$$

For clarity, the minimum of the finitely many quadratic Taylor terms is exactly (15); taking a minimum cannot increase the maximal absolute remainder. Also, with $z = x/\sqrt{t}$,

$$\partial_{tt}u_t(x) = \frac{K}{4t^2}(3z - z^3/c^2)e^{-z^2/(2c^2)}, \quad \|\partial_{tt}u_t\|_\infty \leq \frac{2Ka}{t^2}.$$

The last bound follows from $\sup_{s \geq 0} se^{-s^2/2} \leq 1$ and $\sup_{s \geq 0} s^3 e^{-s^2/2} \leq 2$. The time Taylor remainder is thus at most Kat^{-2} . For $t \geq 1$,

$$\|Tu_t - u_{t+1}\|_\infty \leq Ct^{-3/2}. \quad (16)$$

The operator T is monotone, commutes with spatial translations and addition of constants, and is nonexpansive in the sup norm. Telescoping (16) therefore gives

$$\|T^n u_\tau - u_{n+\tau}\|_\infty \leq C \sum_{j=0}^{n-1} (\tau + j)^{-3/2} \leq \frac{3C}{\sqrt{\tau}}. \quad (17)$$

Finally, the definition of η implies the pointwise initial bounds

$$u_\tau(x - R) - \eta \leq \mathbf{1}_{\{x \geq 0\}} \leq u_\tau(x + R) + \eta.$$

Apply T^n , use its three invariances, and then (17). This proves (14). $\square$

Corollary 11 (CDF limit when $b > 0$). *For each $x \in \mathbb{R}$, $G_n(x\sqrt{n}) \rightarrow F(x)$.*

Proof. Fix τ, R and substitute $x\sqrt{n}$ in (14). Both shifted arguments tend to x , so the limsup and liminf differ from $F(x)$ by at most $\eta + 3C/\sqrt{\tau}$. First choose τ large and then R large. The error tends to zero. For explicit choices at tolerance $0 < \epsilon < 1$, take

$$\tau \geq \max\{1, (6C/\epsilon)^2\}, \quad R = a\sqrt{2\tau \ln(4/\epsilon)}.$$

The Gaussian Chernoff bound gives $\eta \leq \epsilon/2$, and the other term is at most $\epsilon/2$. These parameters are fixed before sending n to infinity, so no uniform convergence over a moving policy grid is being assumed. $\square$

6 Zero variance and convergence of the means

Lemma 12 (The degenerate endpoint). *If $a > 0$ and $b = 0$, then*

$$G_n(x\sqrt{n}) \longrightarrow F_{a,0}(x) = \begin{cases} 0, & x \leq 0, \\ 2\Phi(x/a) - 1, & x > 0. \end{cases}$$

Proof. This step concerns the centered increment control problem only. For $\varepsilon > 0$ add an independent Rademacher increment $\varepsilon\xi_t$ at each time and for every action. Retain both the original output and ξ_t as observed labels. The new centered increments have extreme variances $a^2 + \varepsilon^2$ and ε^2 ; write G_n^ε for their Bellman CDF. Extra labels do not change the Bellman recursion: conditional on the history the next increment law depends only on the action, and the remaining terminal threshold depends only on the current sum.

The perturbed and unperturbed sums under the same action choices differ by $\varepsilon \sum_{t=1}^n \xi_t$. For every $d > 0$, Chebyshev gives

$$\mathbb{P} \left(\left| \varepsilon \sum_{t=1}^n \xi_t \right| > d\sqrt{n} \right) \leq \varepsilon^2/d^2,$$

uniformly over policies. Every original policy is a perturbed policy that ignores the extra coins. Conversely, an original controller can emulate a perturbed policy by drawing those coins as private randomness. The optimal original value is no larger than that randomized policy's value. These two observations give

$$G_n^\varepsilon((x-d)\sqrt{n}) - \varepsilon^2/d^2 \leq G_n(x\sqrt{n}) \leq G_n^\varepsilon((x+d)\sqrt{n}) + \varepsilon^2/d^2. \quad (18)$$

Apply Corollary 11 with $a_\varepsilon = \sqrt{a^2 + \varepsilon^2}$ and $b_\varepsilon = \varepsilon$. Then let $\varepsilon \downarrow 0$ with $d = \sqrt{\varepsilon}$. The explicit CDFs (13) converge to the continuous CDF $F_{a,0}$, also at the moving arguments $x \pm \sqrt{\varepsilon}$. At $x = 0$, both limits are zero directly from the two formulas. Equation (18) proves the claim. $\square$

Lemma 13 (Mean limit). *For $a \geq b \geq 0$,*

$$\lim_{n \rightarrow \infty} \frac{e_n}{\sqrt{n}} = \sqrt{\frac{2}{\pi}} (a - b). \quad (19)$$

Proof. Let $B = \max_{A,y} |-\log_2 P_A(y) - h|$. If $B = 0$ the claim is immediate; assume $B > 0$. For any policy the centered increments have conditional mean zero and absolute value at most B . Convexity bounds their conditional exponential moment by $\cosh(\lambda B) \leq e^{\lambda^2 B^2/2}$. Iteration and exponential Markov therefore give, uniformly over all policies,

$$\mathbb{P}(M_{\pi,n} > t\sqrt{n}), \quad \mathbb{P}(M_{\pi,n} \leq -t\sqrt{n}) \leq e^{-t^2/(2B^2)} \quad (t > 0).$$

By (4), both tails of $Z_n/\sqrt{n}$ have these same bounds. Hence the signed tail formula

$$\frac{e_n}{\sqrt{n}} = \int_0^\infty [1 - G_n(t\sqrt{n})] dt - \int_0^\infty G_n(-t\sqrt{n}) dt$$

admits dominated convergence, using Corollary 11 or Lemma 12. When $b > 0$, its limit is

$$\begin{aligned} \int_0^\infty [1 - F(t)] dt - \int_0^\infty F(-t) dt &= \frac{2a^2}{(a+b)\sqrt{2\pi}} - \frac{2b^2}{(a+b)\sqrt{2\pi}} \\ &= \sqrt{2/\pi} (a - b). \end{aligned}$$

Here $\int_0^\infty [1 - \Phi(t/c)] dt = c/\sqrt{2\pi}$. For $b = 0 < a$ the negative integral vanishes and the positive one is $a\sqrt{2/\pi}$. If $a = b = 0$, all increment laws are point masses at zero and $G_n = G_0$, so the assertion is immediate. $\square$

Proof of Theorem 1. Lemma 2 and Lemma 13 give

$$\frac{C_n^{\text{causal}} - nh}{\sqrt{n}} = \frac{e_n}{\sqrt{n}} + O(n^{-1/2}) \longrightarrow \sqrt{2/\pi} (a - b).$$

If all varentropies vanish, all laws are uniform. Equal entropy means their supports have the same size k . A single uniform seed on $\{1, \dots, k\}^n$, with the appropriate output relabeling for each requested action, has entropy $n \log_2 k = nh$. The all- P transcript gives the reverse inequality. The stronger bound (2) is proved in Section 8. $\square$

7 Rigidity of near-optimal sources and higher moments

Let $W_{a,b}$ have CDF (13) when $b > 0$, the half-normal CDF of Lemma 12 when $b = 0 < a$, and the point mass at zero when $a = b = 0$. In this section Z_n is the *centered* Bellman variable defined in (3); the uncentered finite-tree envelope of Theorem 3 is $nh + Z_n$.

Lemma 14 (Envelope convergence in transport). *The laws of $Z_n/\sqrt{n}$ converge to $\mathcal{L}(W_{a,b})$ in W_1 . They also converge in every fixed absolute moment.*

Proof. The preceding CDF limit gives weak convergence. Lemma 13 bounds both tails of $Z_n/\sqrt{n}$ by $e^{-t^2/(2B^2)}$, uniformly in n . For any $r > 0$, the formula $\mathbb{E}|X|^r = r \int_0^\infty t^{r-1} \mathbb{P}(|X| > t) dt$ and dominated convergence give absolute-moment convergence. For completeness, the one-dimensional quantile coupling identity is

$$W_1(\mathcal{L}(X), \mathcal{L}(Y)) = \int_{\mathbb{R}} |F_X(t) - F_Y(t)| dt.$$

It follows by integrating the lengths of the intervals between the two quantiles over their common uniform parameter (Tonelli's theorem). The Gaussian tail majorant, together with pointwise CDF convergence, therefore proves the W_1 statement directly. The deterministic case $B = 0$ is immediate. $\square$

Theorem 15 (Information-spectrum rigidity). *Let S_n be any sequence of exact causal seeds for the fixed finite family of equal-entropy laws in Theorem 1. If*

$$H(S_n) - C_n^{\text{causal}} = o(\sqrt{n}),$$

then, writing $J_n = -\log_2 \mathbb{P}(S_n)$,

$$\mathcal{L}\left(\frac{J_n - nh}{\sqrt{n}}\right) \longrightarrow \mathcal{L}(W_{a,b}) \quad \text{in } W_1. \quad (20)$$

In particular this holds for entropy minimizers and for the greedy seeds.

Proof. Apply Theorem 9 to the uncentered envelope $nh + Z_n$, center nh , and scale $\sqrt{n}$, using Lemma 14. Equivalently, the exact finite-horizon distance is

$$W_1\left(\mathcal{L}\left(\frac{J_n - nh}{\sqrt{n}}\right), \mathcal{L}\left(\frac{Z_n}{\sqrt{n}}\right)\right) = \frac{H(S_n) - nh - e_n}{\sqrt{n}} \leq \frac{H(S_n) - C_n^{\text{causal}} + \log_2 e}{\sqrt{n}}.$$

The triangle inequality finishes the proof. $\square$

For $a = b = \sigma > 0$, $W_{a,b}$ is exactly $N(0, \sigma^2)$. For $b = 0 < a$, it is the law of $a|N(0, 1)|$. For $a > b > 0$, its density is continuous at zero and consists of Gaussian pieces with different scales, as obtained by differentiating (13). The assertion concerns the information law, not uniqueness of the seed alphabet or of its deterministic decoders.

Theorem 16 (All polynomial moments for the greedy source). *For the greedy information J_n^G and every $r > 0$,*

$$\mathbb{E}\left|\frac{J_n^G - nh}{\sqrt{n}}\right|^r \longrightarrow \mathbb{E}|W_{a,b}|^r.$$

When $a + b > 0$,

$$\mathbb{E}|W_{a,b}|^r = \frac{2^{r/2} \Gamma((r+1)/2)}{\sqrt{\pi}(a+b)} (a^{r+1} + b^{r+1}). \quad (21)$$

In particular,

$$\frac{\text{Var}(J_n^G)}{n} \longrightarrow a^2 - ab + b^2 - \frac{2}{\pi}(a-b)^2. \quad (22)$$

The right side is zero when $a = b = 0$.

Proof. Write $Y_n = (J_n^G - nh)/\sqrt{n}$. The stochastic sandwich implies, for $t > 0$,

$$\begin{aligned}\mathbb{P}(Y_n < -t) &\leq e^{-t^2/(2B^2)}, \\ \mathbb{P}(Y_n > t) &\leq \mathbb{P}(Z_n/\sqrt{n} > t/2) + \mathbb{P}(E > t\sqrt{n}/2) \\ &\leq e^{-t^2/(8B^2)} + e^{-(\ln 2)t/2}, \quad n \geq 1.\end{aligned}$$

These bounds have all polynomial moments and are uniform in n . Combine them with Theorem 15 and the tail integral formula. For the moment formula, integrate the density on each half-line and use

$$\int_0^\infty x^r e^{-x^2/(2c^2)} dx = 2^{(r-1)/2} c^{r+1} \Gamma((r+1)/2).$$

The same formula holds at $b = 0$ by the half-normal density. The mean is $\sqrt{2/\pi}(a - b)$ and the second moment is $(a^3 + b^3)/(a + b) = a^2 - ab + b^2$. Subtract its squared mean. If $B = 0$, the exact uniform construction in Theorem 1 is also the greedy construction: all extracted weights are 2^{-nh} . $\square$

Proposition 17 (Near-optimal entropy does not control variance). *Even within a fixed finite equal-entropy model, there exist exact seeds S'_n with $H(S'_n) - C_n^{\text{causal}} = O(1)$ but $\text{Var}(-\log_2 \mathbb{P}(S'_n))/n \rightarrow \infty$.*

Proof. Append to a greedy seed an independent random variable R_n , ignored by the decoder. For $n \geq 2$, let $p_n = n^{-2}$, $M_n = 2^{n^2}$, and give R_n one atom of probability $1 - p_n$ and M_n other atoms, each of probability p_n/M_n . Then

$$H(R_n) = h_2(p_n) + p_n \log_2 M_n = h_2(n^{-2}) + 1 = O(1).$$

Its information has two possible values, and their difference is $n^2 + 2 \log_2 n + \log_2(1 - n^{-2})$. Thus

$$\text{Var}(-\log_2 \mathbb{P}(R_n)) = p_n(1 - p_n)(n^2 + 2 \log_2 n + \log_2(1 - n^{-2}))^2 \sim n^2.$$

For $S'_n = (S_n^G, R_n)$ the entropies and information variances add, by independence. The greedy entropy gap is at most $\log_2 e$. The claimed entropy bound and divergent normalized variance follow. Each appended alphabet is finite. This does not contradict W_1 rigidity: first moments are controlled, but second moments need not be. $\square$

Theorem 18 (Optimal fixed-confidence information thresholds). *For $0 < p < 1$ define $Q_n(p) = \inf_S q_p(J_S)$, where the infimum ranges over exact causal seeds. Then*

$$Q_n(p) = nh + \sqrt{n} q_p(W_{a,b}) + o(\sqrt{n}).$$

The greedy seed attains this asymptotic for every fixed p simultaneously. For $b > 0$, its explicit coefficient is

$$q_p(W_{a,b}) = \begin{cases} b \Phi^{-1}\left(\frac{(a+b)p}{2b}\right), & p \leq b/(a+b), \\ a \Phi^{-1}\left(1 - \frac{(a+b)(1-p)}{2a}\right), & p \geq b/(a+b). \end{cases}$$

For $b = 0 < a$, it is $a\Phi^{-1}((1+p)/2)$; for $a = b = 0$, it is zero.

Proof. Every exact seed satisfies $nh + q_p(Z_n) \leq q_p(J_S)$, while the greedy seed is feasible, so $nh + q_p(Z_n) \leq Q_n(p) \leq q_p(J_n^G)$. At every $p \in (0, 1)$ the nondegenerate limiting CDF is continuous and strictly increasing in a neighborhood of its p -quantile. The CDF limits imply convergence of both normalized bounding quantiles: for any $\epsilon > 0$, the limiting CDF values at $q_p \pm \epsilon$ lie on opposite sides of p , and eventually so do the approximating CDFs. This proves the assertion by squeezing. Invert (13) to obtain the displayed expressions. The deterministic case uses the exact uniform seed. These are information quantiles, not a claim about exact fixed-length random-bit generation of arbitrary irrational probabilities. $\square$

8 Equal positive varentropies: the logarithmic upper bound

The main second-order argument above is self-contained. For the sharper remainder when $a = b = \sigma > 0$, we use the following specialization of Fan–Grama–Liu, Theorem 2.1 [11].

External input (with its exact hypotheses). Let $(\xi_i, \mathcal{F}_i)$ be martingale differences such that, almost surely,

$$\sum_{i=1}^n \mathbb{E}(\xi_i^2 \mid \mathcal{F}_{i-1}) = 1, \quad \left| \mathbb{E}(\xi_i^k \mid \mathcal{F}_{i-1}) \right| \leq \frac{1}{2} k! \epsilon^{k-2} \mathbb{E}(\xi_i^2 \mid \mathcal{F}_{i-1}) \quad (k \geq 2),$$

where $0 < \epsilon \leq 1/2$. There is an absolute constant C_{FGL} such that, for all real x ,

$$\left| \mathbb{P} \left(\sum_i \xi_i \leq x \right) - \Phi(x) \right| \leq C_{\text{FGL}} (1 + x^2) \epsilon |\ln \epsilon| \exp \left(-\frac{\hat{x}_\epsilon^2}{2} \right), \quad \hat{x}_\epsilon = \frac{2|x|}{1 + \sqrt{1 + 2\epsilon|x|}}. \quad (23)$$

This is that theorem with its quadratic-variation error parameter $\delta = 0$.

Proof of (2). Put $B = \max_{A,y} |-\log_2 P_A(y) - h|$. For every deterministic policy use

$$\xi_i = \frac{D_{A_i}(Y_i)}{\sigma\sqrt{n}}, \quad \epsilon_n = \frac{B}{\sigma\sqrt{n}}.$$

The conditional variance of every ξ_i is exactly $1/n$, regardless of which action the policy chooses. Also $|\xi_i| \leq \epsilon_n$, so

$$\left| \mathbb{E}(\xi_i^k \mid \mathcal{F}_{i-1}) \right| \leq \epsilon_n^{k-2} \mathbb{E}(\xi_i^2 \mid \mathcal{F}_{i-1}) \leq \frac{1}{2} k! \epsilon_n^{k-2} \mathbb{E}(\xi_i^2 \mid \mathcal{F}_{i-1}).$$

Thus all the hypotheses hold once $n \geq 4B^2/\sigma^2$, with identical parameters for all policies. Taking the infimum of their CDFs preserves the same two-sided error about Φ . Consequently

$$|G_n(\sigma\sqrt{n}x) - \Phi(x)| \leq C_{\text{FGL}} (1 + x^2) \epsilon_n |\ln \epsilon_n| e^{-\hat{x}_{\epsilon_n}^2/2}. \quad (24)$$

This is a pointwise uniform bound, not merely a separate Wasserstein estimate for each policy.

The right side has an integrable envelope independent of $\epsilon_n \leq 1/2$. Indeed, for $|x| \geq 1$,

$$1 + \sqrt{1 + 2\epsilon_n|x|} \leq 1 + \sqrt{1 + |x|} \leq 3\sqrt{|x|}, \quad \hat{x}_{\epsilon_n}^2/2 \geq 2|x|/9.$$

It follows that

$$\int_{\mathbb{R}} (1 + x^2) e^{-\hat{x}_{\epsilon_n}^2/2} dx \leq \frac{8}{3} + 2 \int_0^\infty (1 + x^2) e^{-2x/9} dx < 400.$$

Since the standard normal has mean zero, the CDF formula for the difference of means and (24) give

$$\begin{aligned} 0 \leq e_n &\leq \sigma\sqrt{n} \int_{\mathbb{R}} |G_n(\sigma\sqrt{n}x) - \Phi(x)| dx \\ &\leq 400 C_{\text{FGL}} B \ln \left(\frac{\sigma\sqrt{n}}{B} \right). \end{aligned}$$

The nonnegativity follows also from stochastic dominance over any single centered policy sum. Lemma 2 now gives, for every integer $n \geq 4B^2/\sigma^2$,

$$0 \leq C_n^{\text{causal}} - nh \leq 400 C_{\text{FGL}} B \ln \left(\frac{\sigma\sqrt{n}}{B} \right) + \log_2 e.$$

Absorbing the finitely many smaller horizons into the constant proves (2). $\square$

9 An explicit error bound for the crash pair

In this section $P = (1/2, 1/4, 1/4)$ and $Q = (q, q, 1 - 2q)$. Let q be the unique root in $(1/3, 1/2)$ of

$$-2q \log_2 q - (1 - 2q) \log_2(1 - 2q) = 3/2.$$

Uniqueness follows since the derivative is $2 \log_2((1 - 2q)/q) < 0$ on this interval, and the endpoint entropies are $\log_2 3$ and 1. Put

$$a = \frac{1}{2}, \quad b = \sqrt{2q(1 - 2q)} \log_2 \frac{q}{1 - 2q}, \quad c = \sqrt{2/\pi} (a - b).$$

These are exact definitions. The accompanying rational computation gives

$$\begin{aligned} 0.410032428181985576080116712465 &< q < 0.410032428181985576080116712466, \\ 0.456450775263963373017231101124 &< b < 0.456450775263963373017231101125, \\ 0.034747254051817864080022137947 &< c < 0.034747254051817864080022137948. \end{aligned}$$

Theorem 19 (Quantitative crash-pair certificate). *For every integer $n \geq 16$,*

$$\left| \frac{C_n^{\text{causal}} - 3n/2}{\sqrt{n}} - c \right| \leq \frac{17}{2} \frac{\sqrt{\ln n}}{n^{1/4}} + \frac{3}{\sqrt{n \ln n}} + \frac{2}{\sqrt{n}}. \quad (25)$$

In particular the unnormalized remainder is $O(n^{1/4} \sqrt{\ln n})$, with the displayed explicit bound.

Proof. Here is a coarse rational certificate for the constants needed in the proof, separate from the high-precision decimal coefficient. The positive logarithm series with the explicit remainder below gives

$$\frac{1500077}{10^6} < H(.410, .410, .180) < \frac{1500078}{10^6}, \quad \frac{1497682}{10^6} < H(.411, .411, .178) < \frac{1497683}{10^6}.$$

Consequently $q_l = .410 < q < q_u = .411$. Integer-power comparisons give

$$\left(\frac{q_l}{1 - 2q_l} \right)^{16} > 2^{19}, \quad \left(\frac{q_u}{1 - 2q_u} \right)^4 < 2^5.$$

Therefore the defining logarithm for b lies between $19/16$ and $5/4$. In particular,

$$\left(\frac{9}{20} \right)^2 < 2q_l(1 - 2q_u) \left(\frac{19}{16} \right)^2 < b^2 < 2q_u(1 - 2q_l) \left(\frac{5}{4} \right)^2 < \frac{1}{4}.$$

For each possible probability p of P or Q , the same rational bracket gives $1/32 < p^2 < 1/2$. Thus $|\log_2 p - 3/2| < 1$, so $B < 1$ in Lemma 13. Since $|D_A|^3 \leq BD_A^2$, it follows that $m_3 \leq 1/4$. Also $\pi > 2$, so $K \leq 20/19$ and

$$\mathcal{C} \leq \frac{20}{19} \left(\frac{1/4}{3(9/20)^2} + \frac{1}{2} \right) = \frac{4430}{4617} < 1.$$

For a given $n \geq 16$ choose

$$\tau = \sqrt{n}, \quad \eta_0 = n^{-1/4}, \quad R = a\sqrt{2\tau \ln(2/\eta_0)}, \quad s = \sqrt{1 + n^{-1/2}}, \quad r = R/\sqrt{n}, \quad T = \sqrt{\ln n}.$$

The Gaussian Chernoff bound and the prefactors at most two in (13) give $\eta \leq \eta_0$. Thus, with $d = \eta + 3\mathcal{C}/\sqrt{\tau} \leq 4n^{-1/4}$, Lemma 10 reads

$$F((x - r)/s) - d \leq G_n(x\sqrt{n}) \leq F((x + r)/s) + d.$$

Let $Y = Z_n/\sqrt{n}$ and let W have CDF F , so $\mathbb{E}W = c$. Integrating the last comparison over $[-T, T]$ gives

$$\left| \int_{-T}^T [F(x/s) - G_n(x\sqrt{n})] dx \right| \leq r + 2Td. \quad (26)$$

To see the constant r explicitly, for any CDF H , $\int_{\mathbb{R}} [H(x) - H(x-r)] dx = r$ when $r \geq 0$. Apply this to $H(x) = F(x/s)$ for the upper signed bound and to $H(x+r) - H(x)$ for the lower one. Each integrand is nonnegative, so restricting its integral to $[-T, T]$ can only decrease it.

For $t > 0$, Lemma 13 and (13) give respectively

$$\mathbb{P}(|Y| > t) \leq 2e^{-t^2/(2B^2)}, \quad \mathbb{P}(|sW| > t) \leq 2e^{-t^2/(2a^2s^2)}.$$

The latter follows by bounding both Gaussian scales by a ; their prefactors sum to two. For any $v > 0$, $\int_T^\infty e^{-t^2/(2v)} dt \leq (v/T)e^{-T^2/(2v)}$. Since $B^2 \leq 1$, $a^2s^2 \leq 1$, and $2B^2 + 2a^2s^2 \leq 3$, the total contribution outside $[-T, T]$ to the absolute signed difference of the means is at most

$$\frac{3}{T}e^{-T^2/2} = \frac{3}{\sqrt{n \ln n}}.$$

Consequently (26) bounds $|\mathbb{E}Y - sc|$ by $r + 2Td + 3/\sqrt{n \ln n}$. Next, $0 \leq (s-1)c \leq 1/(2\sqrt{n})$, using $c < 1$ and $\sqrt{1+u} - 1 \leq u/2$. For $n \geq 16$, $\ln 2 \leq \frac{1}{4} \ln n$, whence

$$r = \frac{1}{2}n^{-1/4}\sqrt{2 \ln(2n^{1/4})} \leq \frac{1}{2}n^{-1/4}\sqrt{\ln n}, \quad 2Td \leq 8n^{-1/4}\sqrt{\ln n}.$$

Finally, Lemma 2 changes $\mathbb{E}Y$ to $(C_n^{\text{causal}} - 3n/2)/\sqrt{n}$ by at most $\log_2 e/\sqrt{n} < 3/(2\sqrt{n})$; the last inequality follows from $\ln 2 > 2/3$. Adding these bounds proves (25). $\square$

n	Certified lower endpoint	Certified upper endpoint
10^8	0.000000000000	0.399831377112
10^{12}	0.000000000000	0.079430259815
10^{16}	0.029587976869	0.039906531235
10^{20}	0.034170431871	0.035324076233
10^{24}	0.034684066372	0.034810441732
10^{32}	0.034746524423	0.034747983681
10^{48}	0.034747253962	0.034747254142

Every row encloses $(C_n^{\text{causal}} - 3n/2)/\sqrt{n}$. Endpoints are rounded outward to twelve decimal places. Nonnegativity is also used for the first two lower endpoints. Already the row $n = 10^{16}$ excludes $c/2 = 0.0173736270259089\dots$; later rows narrow around c .

Reproduction and arithmetic certificate. Run `python3 certify_crash_pair.py`. Only the Python standard library is required. The output JSON records the coefficient intervals, all table rows, and the status `PASS_EXACT_RATIONAL_ARITHMETIC`. Logarithms are enclosed with 64 terms of the positive series

$$\ln x = 2 \sum_{j=0}^{63} \frac{z^{2j+1}}{2j+1} + R, \quad z = \frac{x-1}{x+1}, \quad 0 \leq R \leq \frac{2z^{129}}{129(1-z^2)} \quad (1 \leq x \leq 2),$$

after powers-of-two range reduction. Entropy signs are checked exactly at both endpoints at every root-bracketing stage. Square roots use integer-square-root enclosures; π uses Machin's identity with alternating-series remainders. Every decimal enclosure is rounded outward. No floating-point value is accepted as an arithmetic premise. The script certifies this arithmetic, not the universal analytic theorem by formal verification. The standalone proof above supplies that theorem.

10 Precise boundaries and remaining questions

All adaptive profiles versus two fixed policies. For just the all- P and all- Q policies, their normalized CDF envelope converges, when $a \geq b > 0$, to

$$F_{\text{fixed}}(x) = \begin{cases} \Phi(x/b), & x \leq 0, \\ \Phi(x/a), & x > 0. \end{cases}$$

Indeed, the ordinary CLT applies to each constant-policy sum, and the minimum of their two CDFs converges to the displayed minimum. The uniform Gaussian tails give convergence of means. Its mean is $(a - b)/\sqrt{2\pi}$, by integrating the two normal tails. The full adaptive envelope has twice that mean. Thus a “half constant” is a property of this restricted policy family, not of every spectrum method. A previously quoted bound of order $O(\sqrt{n})$ from a “Theorem 7” is compatible with identifying a sharp coefficient of $\sqrt{n}$. That earlier theorem’s complete text is not available here, so this note does not certify a stronger comparison with it.

The logarithmic regime is not classified. For equal positive varentropies we prove an $O(\log(n + 1))$ upper bound, not a matching lower bound or a logarithmic coefficient. A universal positive lower bound cannot hold: for $P = Q$, the seed consisting of n independent P samples has entropy nh , and one transcript gives the reverse bound. More generally this is true when the action laws are relabelings of a common law. Which genuinely distinct pairs have logarithmic excess remains a question. A difference of third cumulants is a candidate diagnostic, not a theorem proved in this manuscript. Uniform control of the adaptive Bellman remainder, rather than an ordinary independent-sum Edgeworth expansion, is needed to settle it. No summability of that nonlinear remainder is asserted here.

Constants and degenerate limits. The crash-pair finite-horizon estimate has explicit constants. The $O(\log n)$ application retains the external constant C_{FGL} . The Bellman constant contains b^{-2} , so its positive- b bound is not uniform as $b \downarrow 0$; the endpoint is proved separately. Likewise the many-action theorem assumes a fixed finite family, not an action family growing with n without uniform hypotheses. We do not give a sharp unequal-entropy classification in this note.

Computation, horizons, and priorities. The algorithm can require exponentially many histories and seed atoms. There is no polynomial-time claim and no assertion of one infinite-horizon consistent finite-entropy source. The finite-tree spectral theorem allows history-dependent kernels, but the explicit square-root coefficient here is proved for fixed memoryless action laws of common entropy. Neither a formal proof-assistant certificate nor independent peer review is asserted. Each literature comparison concerns the stated models and verified primary-source claims; it is not a proof of absence of other prior work.

11 Reproducibility and a small exact optimum

The accompanying standalone Python files use rational arithmetic for finite-tree reconstruction and certificate checks. For

$$P = (1/2, \underbrace{1/32, \dots, 1/32}_{16}), \quad Q = (\underbrace{1/4, 1/4}_2, \underbrace{1/16, \dots, 1/16}_8),$$

the entropy is 3 for both actions and the centered information laws are respectively $\{-2, 2\}$ and $\{-1, 1\}$, each equiprobable. At horizon two, the Bellman CDF has masses $1/4$ at centered information levels $-2, 0, 1, 4$. Its mean is $3/4$ and hence every exact seed has entropy at least

$6 + 3/4 = 27/4$. The greedy algorithm constructs a seed with four atoms of weight $1/16$, sixteen of weight $1/64$, thirty-two of weight $1/128$, and 256 of weight $1/1024$. Reconstruction of every history is checked by the independent verifier; its entropy is $(4 + 6 + 7 + 10)/4 = 27/4$. Thus this finite instance has exact optimum $27/4$. The code exports the actual response maps, not just these weights.

The broader verifier enumerates all 128 realized adaptive policies in a binary depth-three example and compares their laws with the constructed seed and its Bellman envelope. The additional verifier checks Renyi moment inequalities, the exact min-entropy endpoint, and the signed stochastic bounds on three finite examples. These are diagnostic certificates; the universal proofs are the arguments in this text. The irrational crash pair is treated by separate rational interval arithmetic, not by replacing it with a nearby rational law.

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